Changing a Networked Mind: Paradigm Dynamics as Structure Learning over a Hidden World

Anonymous Submission

Anonymous Institution(s)

Abstract. How does a scientific community change its mind—not about a fact, but about the *frame* its facts live in? We model a paradigm as a directed network of commitments and a community as bounded active-inference agents who must *learn* that network's structure while valuing some conclusions over others and pooling beliefs with peers. Built from its field's default choices—Gaussian beliefs, precision-weighted pooling—such a community provably cannot run the Kuhnian cycle: conflict resolves into confident compromise, any contact forces consensus, and the lone discoverer of a new dimension is averaged away. We then add the one quantity whose absence those theorems name: *reliability, inferred from surprise*—each agent trusts what does not keep surprising it, applying the same Student-t weight $\lambda = (\nu + 1)/(\nu + z^2)$ to its own sensory channels and to its peers. This single gate generates the missing phenomenology endogenously: the community doubts its instruments before its theory (denial), holding the old paradigm at a fraction of the un-gated conversion rate and rehabilitating the channels it condemned; conflict registers as suspended confidence rather than growing certainty; communities sever trust, persist as schools past a sharp incommensurability threshold set by the confidence each accumulated apart, and re-merge through a distrust–plateau–rehabilitation arc whose ending, unlike its onset, is fast—the reopening gate accelerates its own reopening. Everything the gate buys is a *timescale*, not a wall: consensus remains the only fixed point, moved out of reach. And a pluralism of frames survives contact only when trust reads the *wiring* of a peer's beliefs, not their opinions. Open-endedness in a community of structure learners is a property of its pooling protocol—designed in, never an emergent grace of scale.

Keywords: Active inference · Structure learning · Belief sharing · Paradigm change · Social epistemology

1 Introduction: changing a frame, not a credence

If we want science to keep going well, goodwill is not enough: we need to know how science *works*—what kind of agent can do science at all, how a community comes to represent a genuinely new dimension of the world, and how it loses information it once held. The question has lately acquired a second audience, because communities of reasoning agents are no longer only studied but *built*—machine-mediated research pipelines, automated literature synthesis, multi-agent learning systems—and every such community commits, in its aggregation step, to an answer it rarely examines: what happens to the member who disagrees. Machine learning has begun to name the property at stake *open-endedness*, the capacity of a system to keep producing artifacts that are novel and learnable, identified both as a grand challenge [28] and as an essential ingredient of any general intelligence [14]; social epistemology has long catalogued its failure modes, echo chambers [23] and premature consensus [32]. What neither supplies is a foundation: a model of a *community* of structure-learning agents explicit enough that one can prove which aggregation dynamics preserve open-endedness and which foreclose it. This paper builds that foundation and gives the question both halves of an answer. The most natural baseline—Gaussian beliefs under precision-weighted pooling—provably lacks the capacity: averaging refines a frame; it cannot grow a new one. And the cheapest repair the failure itself names—making each source's *reliability* a quantity inferred from surprise rather than a constant—restores the missing dynamics wholesale, at a price the model makes precise: every protection it buys is a timescale, never a wall.

To change one's mind about a fact is to update a belief; to change a paradigm is to change the *frame* in which beliefs are posed. A frame is not a credence but a *structure*: a network of conditional dependencies through which information propagates. A community that revised only its credences, leaving the network fixed, could never change frame at all. This is the phenomenon Kuhn named but could not run forward [19]: nothing in *The Structure of Scientific Revolutions* says, mechanistically, why a community holds a structure of commitments long past the first disconfirming evidence, nor what finally makes the holding give way.

When the chemical community gave up phlogiston it did not revise a single credence. It rewrote a network of commitments in which combustion, calcination, respiration, and the theory of acids were all load-bearing on a hub that had to go: late-phlogistic theorists proposed cascading revisions that proved mutually incompatible [40, 43], defending less a hypothesis about combustion than a constellation of entangled commitments [42]. Phlogiston was load-bearing for too much.

Network epistemology, the dominant formalism for such communities, cannot represent this case. Its agents hold a scalar credence in a fixed proposition and update it on a trust graph [24, 32, 75]—a programme with striking results on transient diversity, polarization, and the division of cognitive labour [16, 17, 29, 45], but the phlogiston puzzle is not a failure to converge on a number. The commitments were *entangled*: revising one forced a re-reading of the others, and the dispute was over which entanglements to keep. Thagard's ECHO [30, 41] captures the structural intuition as undirected coherence, but lacks the directed dependencies, the carry-over cost of mind-change, and the expand/reduce asymmetry we derive below. Active-inference accounts supply an endogenous, precision-based lock-in mechanism [1] and warn that naive aggregation degrades collective precision [47, 48]; the variational cost of mind-change has been formalized for a single agent as a motivated trade between revision cost and the utility of a held belief [15]. What none of these does is make *belief structure itself* the object of inference. A scalar credence over a fixed menu of hypotheses cannot say how revising one commitment forces a re-reading of the rest, nor how a community arrives at a *new* way of carving up the world: its members can herd on a number or split over it, but they can never come to hold different *kinds* of world—only different values on a shared one.

This paper lets a piece of *structure* travel, so that one agent can come to represent a commitment its peers do not. Its agents carry a directed network of commitments and learn that network by Bayesian model expansion and reduction; they are *bounded*, sampling a complex world through narrow channels, valuing some conclusions over others, and judging what they receive against their own model. Section 3 assembles this community from the field's own default choices, states the two classical theorems under which those defaults can neither reject a conflicting source nor sustain a school—so that *conflict resolves into confident compromise*, and the Kuhnian cycle provably cannot run—and then equips every agent with the one inference the theorems forbid the defaults: reliability, read off surprise, applied alike to the agent's own instruments and to its neighbours. Section 4 drives the gated community through the full cycle—normal science, anomaly, denial, crisis, discovery, schism, rehabilitation—and measures what one inferred quantity buys and what it still cannot. The story, compactly, is about averaging—what belief revision is when value enters as a tilt rather than a stiffness, what propagation-by-averaging does to a population of structure learners, and what it takes for multi-agent structure learning to be *open-ended* rather than self-sealing.

2 What a model of science must contain

Three things, at least. *Structure as the object of inference*: a scientist confronting a new domain is not given the right variables, let alone the dependencies among them—whether two regularities share a hidden common cause, are lawfully coupled, or merely coincide is exactly what she must learn. Frames differ in kind, not in degree, so credences over a fixed menu cannot carry them; the theory-acquisition literature reaches the same conclusion, casting conceptual development as a stochastic search over structures [34, 35, 49]. Once structure is the object, exploration stops being a hand-tuned bonus—an ϵ, a temperature, a switching cost [24, 32]—and becomes the epistemic value of *expanding* the model to entertain a commitment not yet in it [9, 10], with Bayesian model reduction as its complement, pruning what the data abandon in closed form [11, 50]. Existing applications of that machinery run on single agents [50, 51] or define "group" as between-subject neuroimaging [52]; a *community* jointly expanding and reducing a paradigm is the open move this paper makes.

Value, because scientists care about their conclusions: a result on which a career rests is held more tightly than its evidence warrants [17, 29], and attitudes demonstrably live on networks of exactly this kind [3]. And *communication*, because science is social: beliefs travel a trust graph and must somehow be combined when they meet. Each requirement has a default formalization, so natural it is hardly ever examined. Beliefs over a network become a precision-weighted Gaussian model; value becomes a utility that tilts the posterior [15]; combination becomes precision-weighted averaging, the information form of classical opinion pooling [60, 62]. And hiding inside the third is a fourth default, so quiet it is rarely stated: the *weights* of the averaging—how much an agent trusts each instrument and each peer—are constants, blind to what the source actually says. Section 3 builds the community out of exactly these defaults, on purpose, and then revokes only the fourth: reliability becomes something an agent *infers*. The question the rest of the paper asks is what that single change buys—whether agents built this way can grow a frame, hold a school, and lose a paradigm, and not merely polish the frame they have.

3 The model

We build the object up from the single agent and then state the population dynamics as an explicit per-round loop (§3.6).

3.1 A paradigm network and its operator

Begin concretely. The phlogiston paradigm, at its smallest, is three commitments wired together: combustion releases phlogiston; calcination is combustion, slowed; respiration is combustion, tamed. Each commitment conditions the next, evidence for any one bears on all, and the question "what if the calx *gains* weight?" has no local answer—it tugs the whole web. A paradigm, in this paper, is that web made explicit.

An agent represents a paradigm as a directed network $G = (V, E, \Pi)$: nodes carry theoretical commitments x_v, edges carry inferential dependencies, and each edge bears a coupling precision π_{uv} measuring how tightly v is bound to u. This is just a generative model whose hidden states are the commitments themselves, with precision playing its usual role: a high-π edge says that belief in u strongly constrains belief in v. Internal coherence is the corresponding precision-weighted graphical model, $p(x \mid G) \propto \prod_{(u \to v) \in E} \psi_{uv}(x_u, x_v)^{\pi_{uv}}$. None of this machinery is ours: it is a linear-Gaussian structural equation model in the lineage of Wright's path analysis [74], edges as path coefficients and precisions as their weights. We work throughout in that *linear-Gaussian* regime, which puts every quantity below in closed form (Appendix A); agents are *honest*—no one misreports, so lock-in is never a dishonesty artifact—and *bounded*, minimizing variational free energy step by step [39] rather than planning over policies.

Revising a commitment is not a local act: changing x_v forces everything downstream to re-equilibrate, along every chain of dependence. If A is the matrix of edge precisions, A^k collects the chains of length k, and summing chains of every length gives $\sum_k A^k = (I - A)^{-1}$—a finite sum on a DAG. Two literatures already own this object: path analysis as the *total-effects matrix* [58, 74], network science as the Katz–Bonacich index [59, 64]; and it is exactly how influence travels between beliefs in precision-weighted Gaussian models [70]. We adopt it as the *propagation operator* $\mathsf{T} = (I - A)^{-1}$, indexed with rows as sources and columns as descendants, so the entry $\mathsf{T}_{iw} = \pi_{i \rightsquigarrow w}$ is the total precision-weighted coupling propagated from i to w along all directed paths (Appendix A).

Conservatism is then recognized rather than posited. Place a unit of weight on every node and ask, for each commitment i, how much total weight a revision of i would disturb. That is the operator applied to the unit source,

$$\kappa \;=\; \mathsf{T}\,\mathbf{1}, \qquad \kappa_i \;=\; 1 + \sum_{w \in \mathrm{desc}(i)} \pi_{i \rightsquigarrow w}, \tag{1}$$

the precision-weighted mass of a node's descendants. In network-science terms this is a centrality score—a node scores high when many strong chains of dependence leave it (formally, the broadcast variant of the Katz index)—and centrality is usually read as influence or status. We read the same number as *revision cost*: a commitment is exactly as expensive to revise as the mass that must re-equilibrate when it moves. Conservatism is thus a centrality the graph already determines, not a disposition assigned by hand.

What we add to this inherited machine comes in two layers. The first is contact with the world. The dense, high-precision interior of the network is the *core*; the sparse periphery in contact with observation is the *belt*, each belt node reading the world through a sensory channel of precision ρ_k. The core carries no direct sensory channel: an interior commitment is revised only by error propagated inward from the belt, through the same operator—costly by Eq. (1) and insulated by position, which is what makes it slow.

The second layer is value, and the operator is run twice. Scientists value some conclusions over others (§2); following the variational-cost account of belief revision [15], we attach to each node an intrinsic utility u_i and let it propagate through the *same* operator: a commitment one wants true lends value to its dependents, so the conviction field is

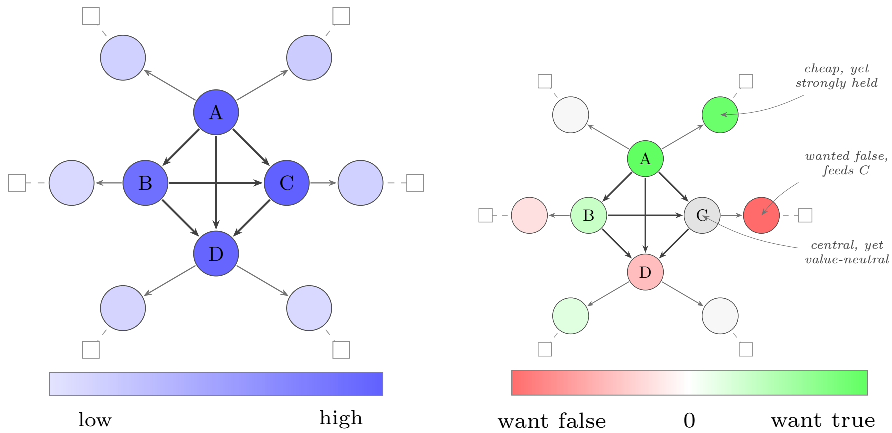
$$U \;=\; \mathsf{T}\,u, \qquad U_i \;=\; u_i + \sum_{w \in \mathrm{desc}(i)} \pi_{i \rightsquigarrow w}\, u_w. \tag{2}$$

In the operator's path-analytic reading, U is the total effect of wanting things true. Where the inherited models propagate a single source, a paradigm carries two: $\mathsf{T}\mathbf{1}$ and $\mathsf{T}u$ are linearly independent unless $u \propto \mathbf{1}$, so the two fields need not align (Fig. 1).

3.2 Motivated revision and the lock-in parameter

The bounded agent revises its belief $q(x)$ over the whole network by trading fidelity to the honest posterior against the value of where the commitments land,

$$\mathcal{F}[q] \;=\; D_{\mathrm{KL}}[q(x) \,\|\, p(x \mid o, G)] - \lambda\, \mathbb{E}_q\!\left[U^{\top}x\right], \qquad \Longrightarrow \qquad q_\lambda(x) \;\propto\; p(x \mid o, G)\, e^{\lambda\, U^{\top}x}, \tag{3}$$



(a) Revision cost κ_i (conservatism) (b) Intrinsic utility u_i (conviction)

Fig. 1: Two fields on one paradigm (schematic). Both panels carry the *same* network—a densely coupled core $\{A, B, C, D\}$ and a sparse belt $\{b_1, \ldots, b_6\}$ in contact with observation. a Revision cost is the carry-over of mind-change, $\kappa = \mathsf{T}\mathbf{1}$ (Eq. 1): a node's cost is the precision-weighted mass of its descendants, a clean radial gradient monotone in centrality. b Intrinsic utility is signed—beliefs one wants true (green), wants false (red), or is neutral toward (white)—and the conviction field $U = \mathsf{T}u$ does not align with κ: C is central yet value-neutral; b_2 is cheap yet strongly held; b_4, which feeds C, is a conclusion the community wants false. The two fields are $\mathsf{T}\mathbf{1}$ and $\mathsf{T}u$, the same operator on two sources, and they are independent vectors.

with λ a single temperature governing how far value may bend the evidence. The operation itself is classical—an exponential tilt of the posterior, the form KL-regularized utility maximization always yields [72, 73]; what is ours is the source of the tilt, a conviction field propagated through the paradigm's own operator. In the linear-Gaussian regime the tilt is a pure mean shift, displacing the commitments by $\lambda \Sigma U$ and leaving the covariance untouched (Appendix A). What opposes the displacement is evidence, also through the mean: a disconfirming channel k with sensory precision ρ_k and read-out H_k pulls the mean back with gain $\propto \rho_k\, H_k \Sigma U$. As conviction drives a disconfirming channel toward $\rho_k \to 0$ that gain vanishes: the channel can no longer move the mean, and the value-displaced commitment stands unrefuted. The fraction of disconfirming sensory precision so silenced,

$$\gamma \;=\; \frac{\sum_{k \in \mathrm{dis}} \rho_k \nvDash [\,\rho_k \text{ gated by the core}\,]}{\sum_{k \in \mathrm{dis}} \rho_k}, \tag{4}$$

is the *lock-in parameter*. In the experiments nothing is silenced by hand: the more a channel's reports threaten what the agent values, the less the agent listens. Concretely, each round channel k carries weight $w_k = \exp(-g\,|U \cdot H_k|)$, where $U \cdot H_k$ measures how directly the channel bears on valued commitments and g sets how strongly caring becomes not-listening.

Note what this mechanism *is*: a representation of which conclusions an agent values, expressed as a displacement of where its beliefs come to rest—and, under forgetting, a displacement that saturates (§3.4) while evidence keeps arriving. A bias, not a resistance. The results will make that distinction load-bearing.

3.3 Structure learning: reduction and expansion

Structure learning revises G itself—the first requirement of §2, that structure be the *object* of inference, now discharged by two asymmetric operations [10, 26]. The asymmetry is the formal counterpart of normal versus revolutionary science.

Reduction is the exact half. Each edge is parameterized by a prior precision α_e ($\alpha_e \to \infty$ pins the edge's parameter to zero; $\alpha_e \to 0$ frees it), the automatic-relevance-determination parameterization of sparse Bayesian learning [20, 31]. Bayesian model reduction [8, 11] scores any pruning in closed form from the already-inverted posterior,

$$\frac{p(o \mid \tilde{M})}{p(o \mid M)} = \mathbb{E}_{p(\theta|o)}\left[\frac{\tilde{p}(\theta)}{p(\theta)}\right], \tag{5}$$

the Savage–Dickey ratio in general form: one inversion scores every candidate reduced structure without re-touching the data.

Expansion reaches what reduction cannot: a commitment no one has represented. Following Friston et al. [10], the agent entertains a new node by wiring it tentatively to existing nodes with free priors ($\alpha_e \to 0$) and letting reduction prune the proposal back to the couplings the data support. Expansion is not closed-form—a new node may enlarge the likelihood's support and requires a real inversion—and this cost asymmetry is why a field reorganizes its periphery continually and its centre only rarely. The agent does not plan *when* to expand: candidates arrive at a Poisson rate, and an arrival is kept only if the model log Bayes factor favours the wired-in node (Appendix D). It must be a rate, because one cannot plan toward a dimension one has not yet conceived—Stanford's problem of unconceived alternatives [27], to which we return. The rate itself, however, is not condemned to be a constant: like every precision in active inference it admits treatment as a *belief*—here, the agent's belief that its own model class is incomplete—and Appendix E gives it that treatment, with the constant rate retained as the agnostic baseline every comparison is run against.

3.4 Forgetting

The introduction asked how a community loses information it once held; forgetting is the model's answer, and it is load-bearing rather than cosmetic. Read once from fixed precisions, $\kappa = \mathsf{T1}$ can only grow as learning adds edges: a paradigm with no forgetting is a ratchet, and lock-in under it is automatic rather than earned. We import the standard remedy unmodified: accumulated precision relaxes toward the prior between updates, $\mathit{\Pi} \leftarrow \mathit{\Pi}_0 + \omega(\mathit{\Pi} - \mathit{\Pi}_0)$ with forgetting factor $\omega \in (0, 1]$ [36, 38], so a deposit k steps old carries weight ω^k. The factor is the agent's prior on the world's structural drift—one forgets at the rate one believes the structure is moving [37]. The operator inherits the time index, $\mathsf{T}_t = (I - A_t)^{-1}$, and a once-central node whose couplings go unreinforced relaxes back toward cheapness. With forgetting, the value tilt saturates at $\lambda U/(1 - \omega)$ and competes with the also-saturated per-step evidence, so conviction holds only above a threshold: an entrenched bloc re-tracks a moving truth at low λ and stays locked at high—motivated persistence turned from a posited bias into a threshold conviction must clear.

3.5 Conflict, and the reliability an agent infers from it

Everything so far is a default, and the defaults have a known pathology: they cannot represent *conflict*. Two classical theorems, one per level of the model, pin it down. *Pooling is a contraction*: precision-weighted averaging on a connected trust graph is DeGroot dynamics in information form [60], and because its weights never depend on the *content* of what is pooled, the spread of beliefs contracts geometrically toward consensus—disagreement can only decay, never regenerate, so a persistent school of thought is impossible by construction [2]. *Gaussian conflict resolves by compromise*: within an agent, a posterior built from light-tailed sources responds to contradiction between them with a confident average—it can never reject a conflicting source, however extreme [4, 5]—and its precision *grows* while doing so [6]: contradiction makes a Gaussian agent more certain, never less. There is no slot in $(\mathit{\Pi}, h)$ for being torn. Together the theorems make the default community a *null regime*: the simplest faithful model of a believing, communicating population, in which crisis, schism, and schools provably cannot arise—only smooth, cumulative refinement of the frame everyone already shares.

The conflict-resolution literature's canonical repair [6] is to stop treating each source's precision as a constant and infer it. We carry the repair as a scale mixture: each channel's noise precision is a latent with its own prior, and its posterior expectation, given the standardized one-step surprise z_k^2 (the squared residual of observation k against the agent's own prediction, in units of its predictive variance), is the classic Student-t weight

$$\lambda_k = \frac{\nu + 1}{\nu + z_k^2}, \tag{6}$$

applied multiplicatively to the channel's evidence deposit. The rule is one sentence: *trust what does not keep surprising you.* A channel reporting on-prediction ($z^2 \sim 1$) keeps full weight; a gross outlier is discounted as ν/z^2—inferred unreliable rather than averaged in—and the single dial ν sets how quickly surprise becomes doubt ($\nu \to \infty$ recovers the Gaussian defaults exactly).

The same functional gates trust *between* agents: $\gamma_{ij} = (\nu_s + 1)/(\nu_s + z_{ij}^2)$, with z_{ij}^2 now the calibrated disagreement between i's and j's posteriors, summed over the commitments both measure—summed, not averaged, because a *localized* disagreement must be able to sever trust; a mean over many agreed-upon nodes would dilute any heresy below the gate's notice. The fusion weights of the pooling step are multiplied by γ, so an agent keeps averaging with neighbours it can predict and quietly stops listening to those it cannot. And because z^2 is just a calibrated gap, the gate can be pointed at different things: at *opinions* (the posterior means above), or at *wiring*—the contested coupling entries of Π itself, a scale-invariant read of how differently two agents structure the world (Appendix C). Where it is pointed will turn out to matter (§4.5).

Note what the gate is not. It adds no objective, no memory, and no new fixed point: λ and γ are instantaneous functions of the current surprise, every weight stays strictly positive, and the contraction theorem still owns the long run—consensus remains the only place the dynamics can rest. Whatever the gate buys, it can only buy as *time*. The results measure how much.

3.6 The population, and the simulation at a glance

The third requirement of §2—communication—enters here, in its default form. A population of N agents inhabits a shared hidden world, each carrying its own paradigm network with its own coupling precisions, conviction field $U = \mathsf{T}u$, and derived conservatism $\kappa = \mathsf{T}\mathbf{1}$. Agents are coupled on a trust graph: communities are blocks of a stochastic block model, with dense intra-community edges and an inter-community coupling weight (*inter*) that we sweep from total isolation to dense contact. Beliefs and accumulated precisions are fused across trust edges in information form—a precision-weighted pooling of posteriors in which an agent's own evidence and its neighbours' carry weight in proportion to trust, the information-form counterpart of trust-weighted averaging in opinion dynamics [55, 60]—and what is not standard is that structural edits spread the same way: an agent whose neighbour has awakened a new node receives the proposal and scores it against its own model by Eq. (5). Figure 2 draws the picture: what travels on the trust graph is not a number but a piece of structure, evaluated against wiring the sender does not see.

Concretely, the experiments of §4 instantiate the phlogiston world: a hub-structured paradigm net in which a single core commitment (phlogiston) is load-bearing for combustion, calcination and respiration; at a fixed step t_{shift} the world's regime flips (the calx is heavier—the gravimetric anomaly), and the truth thereafter contains a dimension (oxygen) that is *not in any agent's network.* One round of the simulation, for every agent:

1. **Observe.** The world emits observations at the belt; channel weights on disconfirming rows are set endogenously by the conviction gate, $w_k = \exp(-g\,|U\cdot H_k|)$ (§3.2), scaled by the agent's gate strength s. With the reliability gate on, each deposit is further weighted by the channel's inferred reliability λ_k (Eq. 6).
2. **Infer.** The agent updates its (value-tilted, Eq. 3) posterior over x by exact linear-Gaussian inference and accumulates precision.

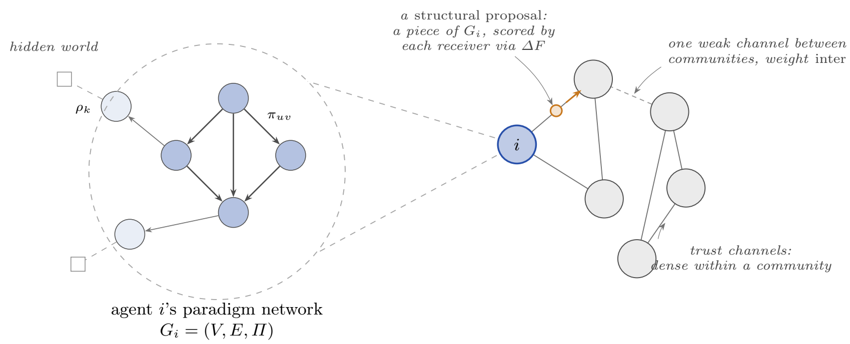


Fig. 2: The population model. Right: the community is a trust graph, every circle an agent. Left, magnified: each agent privately holds a paradigm network $G_i = (V, E, \Pi)$—commitments coupled by precisions π_{uv}, a belt reading the hidden world through sensory channels of precision ρ_k—from which its conservatism $\kappa = \mathsf{T}\mathbf{1}$ and conviction $U = \mathsf{T}u$ are derived. What passes along trust channels (amber) is a *structural proposal*: a piece of paradigm, e.g. a newly awakened node, which each receiver scores against its own network by ΔF (Eq. 5); posteriors fuse precision-weighted along the same channels. The trust graph is fixed at the start (a stochastic block model): channels are dense and strong inside a community, and a single weak weight *inter*—the main dial of §4—connects communities, so what reaches an agent from outside is its neighbours' evidence and proposals, scaled by trust.

3. **Crisis check (reduction).** Each candidate prune is scored on the move ledger: the closed-form prune evidence ΔF (Eq. 5) plus λ_i times the equally closed-form conviction-value change of the prune, $\Delta U = U^{\top}(\mu_{\text{reduced}} - \mu)$, read from the reduced posterior the Savage–Dickey identity already exhibits. Both protective belt edges scoring $\Delta F + \lambda_i \, \Delta U > 0$ is a *crisis*, and the agent applies the reduction—the hub is released (Appendix D).
4. **Discovery check (expansion).** After reduction, the residual the old paradigm absorbed becomes visible as a coherent pattern in prediction error; when a Poisson proposal arrives (rate r), its *direction* is read from that pattern's leading eigenvector, and the wake is accepted iff the model log Bayes factor is positive—the data must hold the new node (§3.3, Appendix D). The rate is constant in the baseline; Appendix E makes it a posterior and a social process, and replaces the one-shot accept with an audited wake-then-prune cycle.
5. **Fuse.** Posteriors and structural edits are pooled across the trust graph, weighted by the intra/inter coupling—and, with the social gate on, by the inferred trust γ_{ij} (§3.5). Whether agents pool over *all* dimensions or only over those they jointly represent is a protocol choice with consequences (Appendix B).
6. **Forget.** Accumulated precision decays by ω (§3.4).

Crisis, revolution and discovery are thus not stipulated events but threshold crossings of the model's own evidence; Appendix D derives both checks from the single tilted objective and lists explicitly what remains a modelling commitment. Table 1 (Appendix B) lists the dials swept; everything else is held fixed, and all code is seeded and ships its own null and positive controls.

One thing deserves emphasis before the results: every ingredient above is its field's default. The belief model is the workhorse of Kalman filtering and structural equations; the value mechanism is the standard formalization of motivated reasoning; the fusion step is classical opinion pooling in information form; expansion and reduction are textbook Bayesian model comparison. Nothing exotic has been smuggled in. The sweep that follows tests what these defaults can do *together*.

4 Results

All results are from the population of §3.6 run on the phlogiston world—normal science, the gravimetric anomaly, crisis, reduction, discovery of oxygen, new normal, every phase a threshold crossing of the agents' own evidence, none stipulated. The nets are hand-wired to expose the geometry, not fitted to data: nothing here is evidence about chemistry, only about how a structured, bounded, valued mind changes. We first state what the default community does, then switch on the gate of §3.5—first within agents, then between them—and follow the cycle through crisis, schism, and rehabilitation.

4.1 The null regime, measured

With every reliability fixed, the sweep (2,488 runs; figures and the full mechanism table in Appendix B) confirms the theorems and locates their boundary. Within an agent, conviction tilts where beliefs settle, never when they yield: value buys a biased resting place, not a delayed reckoning [15], and its one causal route to lock-in runs through gating one's own disconfirming channels. At population scale that protection loses to the faintest connection: at $inter = 0.005$—trust toward outsiders two hundred times weaker than trust within the community—92% of the dogmatic community converts, however dogmatic, because fusion shares *evidence*, not conclusions: an agent can silence its own instruments, not its neighbours' (Fig. 9). Between agents, pooling kills the lone discoverer—a concept woken by one agent at a time is averaged away by peers whose priors read *has no such variable* as *certainly zero*—and every mechanism we layered on the baseline (adequacy-driven exploration, social excitation, audited curiosity, value accretion) either re-times the passage to consensus or synchronizes the population for it; none lets two connected communities holding the same data remain in disagreement (Table 2). The null regime is normal science, and nothing in it can do anything else.

4.2 Within an agent: conflict becomes crisis, and points to its own cause

The smallest possible setting already shows everything the channel gate changes. One agent holds a three-commitment chain $A \to B \to C$; at step 30 channel A turns persistently contradictory—the world keeps reporting a value the rest of the agent's network cannot reconcile. The Gaussian baseline does what the theorem says it must: it settles confidently between the channel's report and what its own couplings imply ($\mu_A = -2.0$, a configuration neither source supports), its variance falling monotonically throughout—the conflict leaves *no trace* in the belief state. The gated agent instead infers the channel unreliable (λ_A falls to 0.16), and its uncertainty does something no light-tailed posterior can: it *rises* while evidence accumulates, peaking eight steps after the conflict begins before recovering (Fig. 3)—crisis as suspended confidence, rendered as a posterior. The conflict also becomes *localizable*: edge strain (Appendix C), a cheap per-step read of how hard each edge fights the agent's own data, ranks the violated coupling above the innocent one at every step, so the Bayes-factor verdict of Eq. (5) need only be paid where the strain points—targeted, rather than scanned, model reduction.

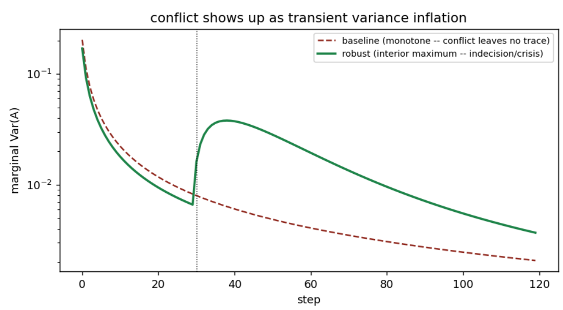


Fig. 3: Conflict as suspended confidence. A channel turns contradictory at step 30 (dotted line). The Gaussian baseline (dashed) grows monotonically more certain—the confident compromise; the gated agent's marginal variance (solid) shows an interior maximum: doubt rises while data accumulate, then resolves as the discounted channel is reorganized around. There is now a slot in the belief state for being torn.

4.3 The community: denial, delay, rehabilitation

Run the same gate in the full population through the regime flip (Fig. 4). At the flip, the anomalous channels become surprising, so the community infers *them* unreliable first: their mean inferred reliability collapses from above 0.85 to 0.17

while the unaffected channels stay fully trusted throughout—selective denial; the instruments are doubted before the theory, which is how the chemists actually received the gravimetric anomaly. The discounted evidence still trickles in, so belief drifts; as it drifts, the anomaly grows less surprising, the gate *rehabilitates* the channels it condemned (final $\bar{\lambda} = 1.27$), and conversion completes—4.3× later than the fixed-precision slide, on the same evidence. The shape is worth stating precisely, because the gate's feedback here is *self-stabilizing* (each agent's surprise falls as its own belief moves): the channel gate *throttles* the conversion—a stretched, decelerating slide, not a stall that breaks—and harsher gates only stretch it further. The sudden half of the Kuhnian shape is real but lives one level up, in the trust gate, whose feedback is *collective*: there, a leak shrinks the inter-camp gap, which reopens the gate, which accelerates the leak—and the merge arrives late and fast (§4.4).

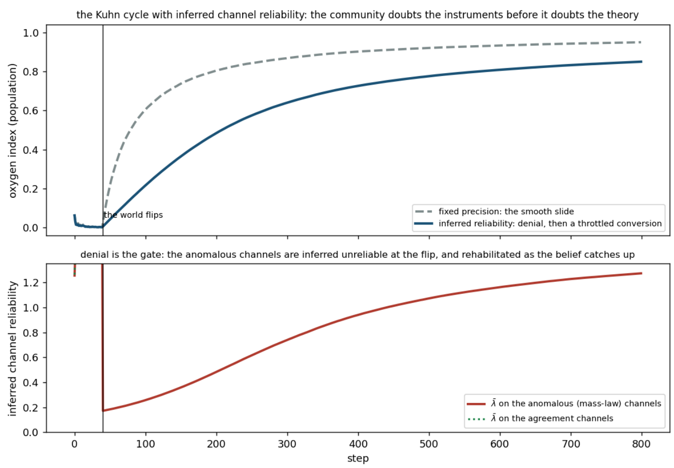


Fig. 4: The Kuhn cycle with inferred channel reliability ($N = 20$, world flips at the vertical line). *Top:* population conversion. Fixed precision (dashed) slides from the flip; the gated community (solid) converts on the same evidence at a fraction of the rate—denial throttles, it does not break. *Bottom:* the mechanism. Mean inferred reliability $\bar{\lambda}$ on the anomalous channels collapses at the flip—denial—while the agreement channels stay trusted; as belief catches up, the channels the community condemned are rehabilitated. Doubt the instruments first, the theory later.

4.4 Between communities: distrust buys time; confidence buys a schism

Point the gate at peers and the contraction theorem's grip loosens—into timescales, exactly as §3.5 promised. The cleanest setting first: two opposed camps on a *complete* graph, no topology to hide behind. Uniform fusion collapses their disagreement to 0.1% of its initial value within a couple of steps; with the gate on, each camp infers the other not worth listening to (cross-camp trust settles at 0.8% of within-camp), and 48% of the initial separation survives the whole horizon. Stable schools, on a connected graph, holding shared data—the configuration the fixed-trust regime forbids—are now *attainable*, as metastable states rather than fixed points.

What decides whether they are *attained* is confidence. Sweeping how much evidence the two camps accumulated before contact (Fig. 5) yields a sharp threshold: below it the gated population merges exactly like the fixed-trust control ($< 1\%$ of disagreement survives); above it most of the disagreement outlives the run (17%/48%/76% as confidence doubles past the threshold), because the gate reads the calibrated gap $z^2 \sim \text{gap}^2 \times \text{precision}$: camps that merely believed differently with more confidence are more mutually surprising. Precision being the integral of exposure time, confidence at contact is what incubation *buys*: incommensurability is not a property of two ideas but of two histories, and it has a clock.

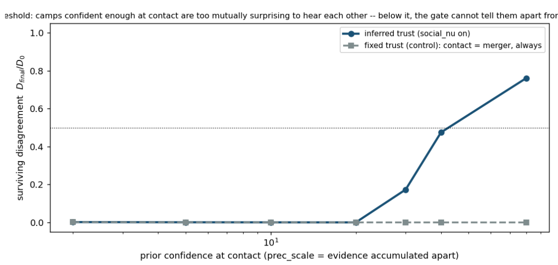


Fig. 5: The schism threshold (two opposed camps, full contact, complete graph). x-axis: prior confidence at contact (evidence accumulated apart); y-axis: fraction of the initial disagreement surviving the horizon. Under fixed trust (squares) contact is merger at every confidence ($\leq 0.2\%$ survives). Under inferred trust (circles) there is a sharp interior threshold: below it the gate cannot tell the camps apart from the control; above it the camps sever trust and most of the disagreement survives. Confidence is the clock of incommensurability.

The full Kuhn cycle puts both halves together (Fig. 6). Re-running the null regime's lock-in boundary with inferred trust, the dogmatic community now *builds its own wall*: as the open community converts, cross-community trust collapses below 1% of within-community trust—self-organized isolation, nothing silenced by hand. But mid-cycle dogmatists have not yet accumulated a schism's worth of confidence, and the gate is memoryless: the evidence that leaks through the still-positive weights moves the laggards, the gap shrinks, the gate *reopens*, and the merge cascades—genuinely late and fast, with the laggards' conversion rate peaking at step 126 of 180, the slow-then-sudden shape the self-stabilizing channel gate of §4.3 cannot produce. Where fixed trust converts the dogmatic community at *inter*$= 0.005$ with $t_{1/2} = 78$ steps, the gated community needs 146–170 and is still incomplete at the horizon (0.54–0.73 converted vs. 0.92); and the trust the gate severed is *rehabilitated* by the merge it failed to prevent (final cross-community trust above 90% of within). Distrust, inferred, has the arc fixed trust cannot draw: distrust → plateau → rehabilitation → merge.

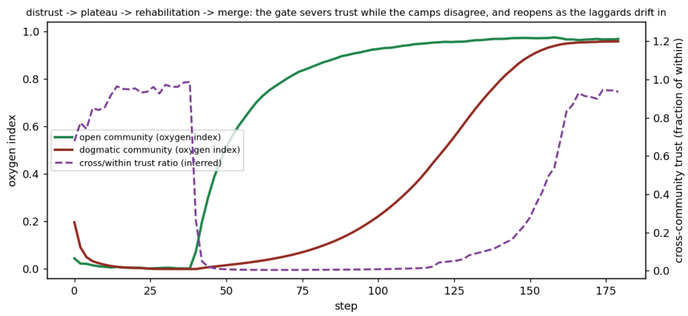


Fig. 6: The rehabilitation arc (full Kuhn cycle, inferred trust, *inter*$= 0.05$). As the open community converts (green) the camps diverge and inferred cross-community trust (dashed, right axis) collapses to ∼1% of within-community trust: the gate severs the channel, and the dogmatic community (red) plateaus at the old paradigm. The leak through the still-positive weights moves the laggards; as they drift in, the gap shrinks, the gate reopens, and the merge completes—restoring the trust whose absence it had inferred. Distrust buys time, not a wall.

4.5 Pluralism lives in the wiring—and only wiring-reading trust can keep it

Everything above gates on *opinions*: the posterior means. The last experiment asks what happens when what divides two communities is not where their beliefs sit but how they are *wired*—the situation the introduction began with. In a world genuinely underdetermined between two structures, two communities, attending differently because they value different conclusions, learn the two different wirings. Connect them under fixed trust and averaging dilutes both into one blurry compromise net (10% of the structural divergence survives). Connect them under the *opinion*-reading gate and exactly the same collapse occurs (10.2%)—the gate never fires, because both camps fit the same blended data and so *agree in their means*: a trust that reads opinions is blind to a pluralism that lives in wiring. Point the same Student-t functional at the contested couplings of Π instead (§3.5), and 94% of the divergence survives full contact, across the whole mixing sweep (Fig. 7). The collective, meanwhile, knows more than its members only as a *union*: averaging never creates an edge no member holds (measured synergy is exactly zero), so the two-wiring knowledge exists in the population precisely as long as the camps do.

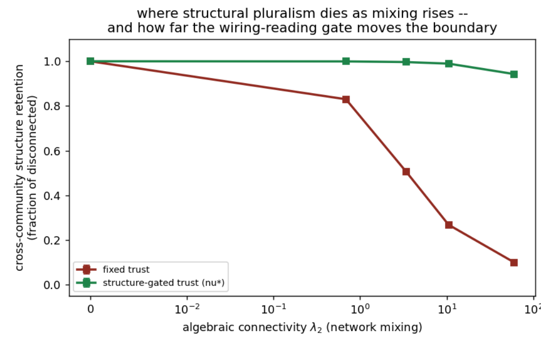


Fig. 7: Where structural pluralism dies as mixing rises. x-axis: algebraic connectivity of the contact graph; y-axis: surviving cross-community structure (fraction of the disconnected benchmark). Fixed trust (red) dilutes both wirings toward one compromise net as mixing rises; the wiring-reading gate (green) preserves the two learned structures at 94% even under dense contact. The opinion-reading gate (not shown) tracks the red curve: the camps agree in means, so it never fires.

One prediction this experiment *refuted*, and the refutation is a scope condition worth stating plainly. We expected a third camp, committed to a wiring the data do not support, to fail to consolidate it. It consolidates it exactly as strongly as the supported camps do: in linear-Gaussian Fisher learning the precision deposit never sees the observation's *value*—attention alone fabricates the coupling pattern, and the data's verdict on an unsupported theory lives in the means, not in $|\Pi|$. Structure-as-precision-pattern, in this model class, is attention-driven: the wiring gate can protect a pluralism, but it cannot tell a supported school from a merely committed one. That discrimination belongs to the means—and to the strain and reliability reads that live on them (§4.2).

Controls. Every experiment ships its own null and positive controls: a world that never shifts produces neither denial nor delay (λ stays above 0.85 on every channel); the fixed-trust arm reproduces the null regime byte-identically (both gates default to off); low-confidence camps merge indistinguishably from the control; and no acceptance bar is tuned anywhere—the wake's sole accept test remains the model log Bayes factor (Appendix D). Configurations, seeds, and the per-experiment headline numbers are tabulated in Appendix C.

5 Discussion

The model's relation to Kuhn can now be stated without allegory, in both directions. The default community is the formalism's *normal-science* limit: smooth, cumulative refinement within a shared frame, with a hard core and protective belt that are derived rather than decreed [66], and provably nothing else. One inferred quantity—reliability, read off surprise and applied alike to instruments and to peers—supplies the revolutionary remainder as *dynamics*: crisis as suspended confidence (the interior variance maximum of §4.2), denial as selective instrument-doubt that throttles conversion (§4.3), the late, fast merge of a reopening trust gate, incommensurability as a confidence threshold with a clock, and schools, schisms, and rehabilitations as metastable states of one trust functional (§4.4). None of these is stipulated; all of them are the Student-t weight of Eq. (6), pointed at different parts of the agent's world.

The recurring price is the result we most want the reader to keep: *everything the gate buys, it buys as time.* The gate is memoryless, its weights stay positive, and consensus remains the only fixed point—so dogma, denial, schism, and school are not different dynamics but the same dynamics on slower clocks, and each protection ends by undoing itself (the trust the gate severed is rehabilitated by the merge it delayed). This is the same shape forgetting gave conviction in §3.4: protection in this model class is always a race between rates, never an absorbing state. Whether *real* communities enjoy any protection stronger than a timescale—or whether incommensurability was only ever patience running out—the model converts from a metaphor into a measurable claim.

For communities of reasoning machines the statement is a design rule. Any aggregation step that averages rather than gates inherits the null regime, whatever the sophistication of its members: it will refine frames and never grow, hold, or split one. Gating restores open-endedness only if it is pointed at the right level—a gate on opinions preserved nothing of a pluralism that lived in wiring (§4.5). Open-endedness in a population of structure learners is a property of the pooling protocol and of *what the protocol reads*, to be designed in; it is not an emergent grace of scale.

Two limits bound the claims, honestly. First, in this model class the precision-pattern is attention-driven: data cannot veto an unsupported wiring (the refuted prediction of §4.5), so the wiring gate protects pluralism without certifying it—the world's verdict stays in the means. Second, everything revolutionary here passes through model expansion over a fixed likelihood, while a genuinely new *instrument* enlarges the likelihood's own support; one cannot place a posterior over the unconceived [27], only over whether one's model has missed something. What a community controls is how hard it looks, what its pooling does to the member who finds it first—and now, how long its distrust is allowed to last.

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A The operator, the tilt, and the Schur residue

Linear-Gaussian construction. The couplings form a structural equation model $x = Ax + \zeta$, $\zeta \sim \mathcal{N}(0, D^{-1})$ with A the weighted DAG adjacency [18]; the choice is for tractability, and a nonlinear coupling would replace T by a local linearization. The propagation operator is the resolvent

$$\mathsf{T} = (I - A)^{-1} = \sum_{k \geq 0} A^k, \qquad \mathsf{T}_{iw} = \sum_{\text{paths } i \rightsquigarrow w} \prod \pi \;=\; \pi_{i \rightsquigarrow w}, \tag{7}$$

the Neumann series finite because a DAG makes A nilpotent. We realise T on the coupling magnitudes $|A|$ so downstream mass does not cancel when a child opposes a parent; the signed A builds the precision matrix. We index T so that κ_i sums row i; under the SEM convention the released code stores the transpose and computes the same fields as column-sums of $\mathsf{T}^\top$, so a reader reproducing $\mathsf{T}\mathbf{1}$ literally against the code's array must transpose first.

The value tilt. With honest posterior $p(x \mid o, G) = \mathcal{N}(\mu, \Sigma)$, the tilted posterior of Eq. (3) is a pure mean shift, $q_\lambda(x) = \mathcal{N}(\mu + \lambda \Sigma U,\, \Sigma)$, with cumulant-generating function $Z(\lambda) = \lambda U^\top \mu + \frac{1}{2}\lambda^2 U^\top \Sigma U$. Note that gating a channel ($\rho_k \to 0$) *raises* the posterior variance and hence Z''; lock-in lives in the vanishing mean-update gain $\rho_k H_k \Sigma U$, not in the curvature.

The Schur residue. Marginalizing a hub condenses its influence into new edges among its neighbours: for a common-cause structure over (x_1, x_2, b),

$$\Pi = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \xrightarrow{\text{marginalize } b} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} - \tfrac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} (1\ 1) = \begin{pmatrix} 1.5 & -0.5 \\ -0.5 & 1.5 \end{pmatrix}, \tag{8}$$

a direct coupling induced where the joint had a zero. Conditioning on a shared *effect* is the dual operation: two commitments that jointly feed one observation are a priori independent, and clamping that observation anti-couples them—explaining away, run on the precision matrix. Both operations fill in the off-diagonal coupling the carry-over cost is built from, and Fig. 8 draws both on the smallest nets that exhibit them.

B The null regime, measured

This appendix carries the quantitative characterization of the default (fixed-reliability) community summarized in §4.1: the full Kuhnian-cycle sweep over the dials of Table 1, 2,488 runs spanning the baseline and every mechanism of Appendix E.

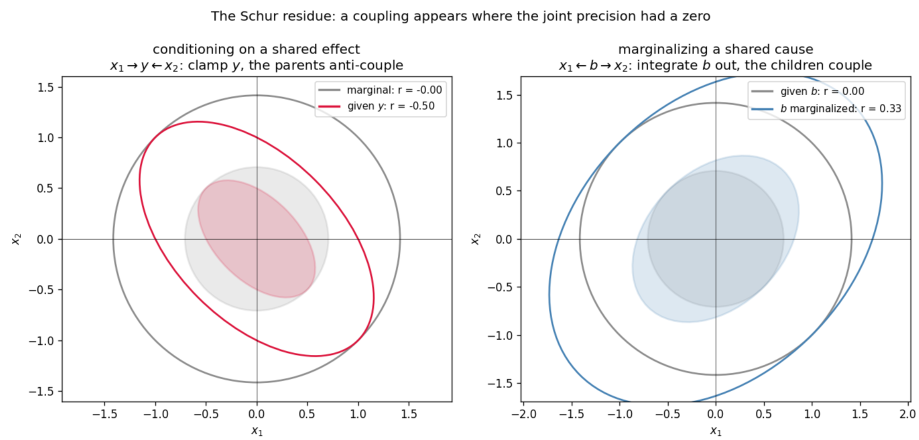

Fig. 8: The Schur residue, computed on the smallest nets that show it. *Left:* conditioning on a shared effect ($x_1 \to y \leftarrow x_2$): the parents are a priori independent (grey, $r = 0$); clamping their common descendant y manufactures an anti-correlation (crimson, $r = -0.50$)—explaining away. *Right:* marginalizing a shared cause (the matrix of Eq. 8): the children are independent given b (grey); integrating b out leaves them coupled (blue, $r = +0.33$). In both cases a coupling appears where the joint precision had a zero—the off-diagonal mass the carry-over cost $\kappa = \mathsf{T1}$ is built from.

Dial	Symbol	What it controls
Social coupling	*inter*	inter-community trust weight (0 = isolation)
Vanguard fraction	—	fraction of agents in the open (low-λ) community
Conviction	λ_d	dogmatic community's value tilt / prune protection
Gate scaling	s	strength of the conviction gate on disconfirming channels
Proposal rate	r (Poisson)	arrival rate of structural proposals
Forgetting	$\omega \in (0, 1]$	decay of accumulated evidence toward the prior
Observation noise	σ_o	sensory noise at the belt
Channel reliability	ν	Student-t gate on the agent's own channels (off = Gaussian)
Social trust	ν_s	Student-t gate on the fusion weights (off = fixed trust)
Fusion protocol	—	pool all dimensions (delta) vs. jointly represented (abstention)
Closed-loop mechanisms	$\beta, \tau, \kappa_r, \lambda_c, \epsilon$	adequacy-driven rate, social excitation, audit credit, value accretion (Appendix

Table 1: The dials swept in the experiments; all other quantities are held fixed. The last two rows are protocol and mechanism variations layered on the baseline; each defaults to "off" (constant rate, one-shot accept, delta pooling, static u, both reliability gates off), reproducing the baseline model exactly.

Only total isolation protects a superseded paradigm. The setup pits a dogmatic community—high conviction, whose agents gate their own disconfirming channels (§3.2)—against an open vanguard already tracking the anomaly, and dials the inter-community trust weight *inter* from 0 (sealed off) to dense contact.

Sealed off, the gating protects: under strong gating ($s = 10$) only 19% ever convert. But at *inter*$= 0.005$, 92% convert to oxygen, however strong the gate and however dogmatic the agents, and the transition is a sharp boundary, not a gradual fade (Fig. 9). A control pins down what does the stalling: keep conviction exactly as high but switch the self-gating off, and the community revolts in full (revolted fraction 1.0 versus 0.0, $\gamma \to 1$). The tilt never stalls anyone (Eq. 4); its one causal route to lock-in runs through the gate, and the gate loses to the faintest connection, because a neighbour's evidence arrives weighted by trust, not by the receiver's conviction. Motivated reasoning in this regime is *wishful*, not *dogmatic*: a bias lives in the potential h; an entrenchment would have to live in the precision Π—which is exactly where the inferred-reliability gate of §3.5 puts it.

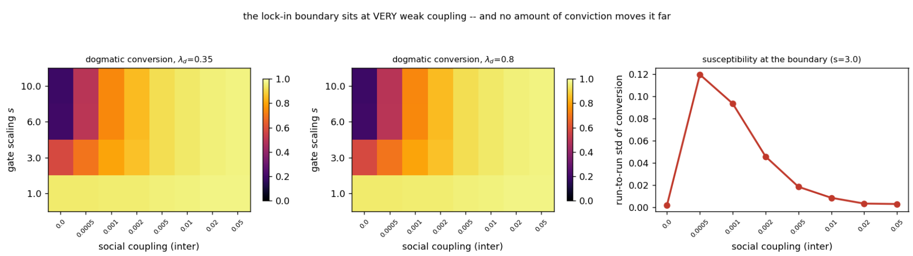

Fig. 9: Only isolation protects a superseded paradigm (phlogiston sweep, fixed trust). *Left, centre:* the fraction of the dogmatic community that converts to oxygen, as a function of contact (*inter*) and of how strongly conviction gates disconfirming channels (s), at two conviction levels ($\lambda_d = 0.35$ and 0.8). Past *inter* ≈ 0.005 nearly everyone converts, and doubling conviction barely changes the picture. *Right:* the run-to-run spread spikes in a narrow band around *inter* $\approx 5 \times 10^{-4}$: a sharp boundary, not a fade.

Pooling kills the lone discoverer—by protocol, not by principle. Discovery means waking a node—oxygen—that no agent's network yet contains; candidates arrive at random (Poisson rate r, §3.3), and the fraction that discovers follows the waiting-time law $1 - (1 - r)^T$. What the rate governs is survival: under the baseline fusion, a discovery made by one agent at a time is averaged away every time, because delta pooling reads *has no such variable*—a fact about an agent's model class—as *maximally certain it is zero*, a belief within its support [47, 62]. The corrected protocol is dimension-aware *abstention* pooling: each entry of the precision matrix is pooled only over the neighbours that represent it. The consequence is a wholesale relocation of the survival threshold (Fig. 10): under delta pooling staggered survival is zero at every rate; under abstention pooling it tracks the discovery fraction itself (0.39/0.64/0.86 at $r_0 = 0.005/0.01/0.02$).

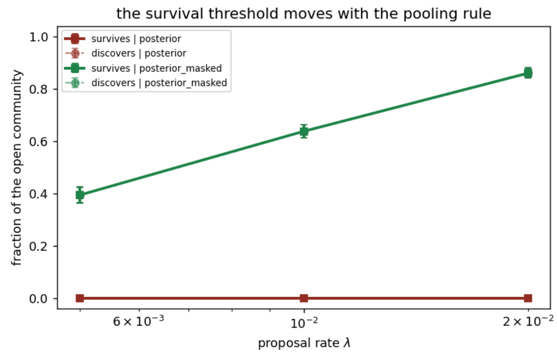

Fig. 10: The survival threshold moves with the pooling protocol (32 seeds per cell). Solid: fraction of the open community whose concept survives; dashed: fraction that discovers at all. Under delta pooling (red) a pinned slot is a confident "no such thing" and staggered survival is zero at every rate; under abstention pooling (green) survival tracks discovery. Incommensurability as a population-level force is a property of the protocol, not of the concepts.

Every escape goes around the averaging, never through it. The sweep's remaining mechanisms sharpen the same conclusion (Table 2; full results in Appendix E). None of them lets two connected communities holding the same data *remain* in disagreement: each either re-times the passage through the consensus bottleneck or synchronizes the population for it.

Escape	What it changes	Measured effect	Verdict
Adequacy-inferred rate	*when* the community looks	crisis-to-discovery 55 $\to$ 29 steps; matched by a constant rate of equal mean	re-times the passage
Social excitation (Hawkes)	how nearly *together* it looks	waking dispersion 64 $\to$ 2 steps; staggered survival 0 $\to$ 0.88–0.95	synchrony outruns the averaging
Audited wake-then-prune	whether curiosity is *safe*	null world: 100% of wakes re-pinned, exactly	keeps the search honest
Value accretion	conviction grown from *inside*	conversion delayed $\sim 2.2\times$, never prevented	buys time, not immunity

Table 2: Four mechanisms layered on the fixed-reliability baseline, none of which opens the consensus engine (Appendix E).

Robustness. The arc is robust across forgetting and observation noise ($\omega \in [0.9, 1.0]$, $\sigma_o \in [0.25, 1.0]$): crisis and reduction survive everywhere—including the no-forgetting wall $\omega = 1$, where only discovery thins: a community that never forgets still sees its paradigm die, but loses the capacity to grow the successor. Each experiment ships its own null and positive controls—a world that never shifts produces neither crisis nor discovery; the gated, isolated community stays locked; Poisson discovery follows the waiting-time law—and no acceptance bar is tuned anywhere: the wake's sole accept test is the model log Bayes factor (Appendix D); whatever the derivations there do not fix is named there as a commitment.

C Inferred reliability: definitions and experiment details

The three reads of one functional. Every gate in the paper is the Student-t scale-mixture weight $\lambda = (\nu + 1)/(\nu + z^2)$—the posterior expectation of a source's latent noise scale given its standardized squared surprise z^2 (one E-step of the EM view of t-regression)—applied to three different surprises. *Channels:* $z_k^2 = (o_k - (H\mu)_k)^2/\mathrm{predvar}_k$ with $\mathrm{predvar}_k = \sigma_o^2 + (H\Pi^{-1}H^\top)_{kk}$, the agent's own surprise at observation k before depositing it; λ_k multiplies the channel's deposit, composing with the conviction gate of §3.2. *Opinions:* $z_{ij}^2 = \sum_v (\mu_j[v] - \mu_i[v])^2/(\Pi_i[v,v]^{-1} + \Pi_j[v,v]^{-1})$, the calibrated disagreement of two posteriors summed over the measured nodes (summed, not averaged: a localized heresy must be able to sever trust); γ_{ij} multiplies the fusion weights, whose row normalization absorbs the overall scale. *Wiring:* the same sum taken over contested coupling entries $x = \Pi[a,c]$, normalized by their pairwise symmetric scale $\max((x_i^2 + x_j^2)/2, \epsilon^2)$—bounded per coupling, and invariant to the global precision growth the forgetting equilibrium produces, so it reads *how* two agents wire the world, not how much evidence they hold. *Edge strain*, the diagnostic of §4.2, is the node-splitting read of the same idea applied to an agent's own edges: the calibrated displacement between the posterior endpoint marginals with and without an edge's deposits, a per-step $O(1)$ read of where the model fights the data that ranks candidate prunes before any Bayes factor is paid.

The experiments. Table 3 lists the configurations; all are seeded, assertion-checked at run time, and write their summary statistics next to their figures.

Scope, stated plainly. Three commitments bound the results. (i) Both gates are *memoryless*: λ and γ are instantaneous functions of current surprise, so every school and schism reported is metastability relative to a horizon, not a new fixed point—the contraction theorem owns $t \to \infty$, and the lock-in rerun shows the trade directly: a harsher gate (ν_s=0.2) holds the boundary longer but leaves the rehabilitation arc unfinished at the same horizon, because the lock and the merge run on the same clock. (ii) The schism control is normalized by the pre-contact stance gap, since one uniform fusion step collapses the fixed-trust camps before the first logged sample. (iii) Structure-as-precision-pattern is attention-driven in the linear-Gaussian class (the Fisher deposit $H^\top \mathrm{diag}(w) H/\sigma^2$ never sees the observation's value), which is what refuted the three-camps prediction of §4.5 and is the right caveat on every wiring-gate claim.

Experiment	Setup	Headline
Conflict crisis (§4.2)	1 agent, chain $A \to B \to C$, channel A contradictory from $t{=}30$; $\nu{=}4$, $T{=}120$	$\lambda_A \to 0.16$; interior variance maximum at $t{=}38$; strain ratio 5.3× on the violated edge
Targeted reduction (§4.2)	1 agent, chain with one world-violated coupling, $T{=}80$	violated edge top-strain at 100% of steps; ΔF verdicts correct from step 1
Crisis trace (§4.3)	$N{=}20$, flip at $t{=}40$, $\nu{=}2$, $T{=}800$	denial $\bar{\lambda}: 0.85 \to 0.17$; stall 4.3×; rehabilitation to 1.27; agreement channels > 0.85 throughout
Gated camps	$N{=}20$, complete graph, opposed camps, $\nu_s{=}0.5$, $T{=}80$	uniform fusion: 0.1% of disagreement survives; gated: 48%; cross/within trust 0.8%
Schism threshold (§4.4)	ditto, sweeping prior confidence at contact	fixed trust $\leq$ 0.2% everywhere; gated $<$ 1% below the threshold, 17/48/76% above
Lock-in rerun (§4.4)	$N{=}80$, full cycle, $\nu_s{=}0.5$, $T{=}180$, *inter* swept	$t_{1/2}$: 78 $\to$ 146–170 at the boundary; conversion 0.54–0.73 vs 0.92; trust $<$ 1% then $>$ 90%
Divergent wiring (§4.5)	$N{=}60$, structurally underdetermined world, two attention camps, $T{=}120$, 5 seeds	retention: fixed 10%, opinion-gate 10.2%, wiring-gate 94%; synergy = 0

Table 3: The gated experiments: configurations and headline numbers.

D The crisis and discovery checks, derived from the ledger

The per-round loop of §3.6 contains two threshold events: the *crisis check* (when does an agent prune the protective edges and release the hub?) and the *discovery check* (when does it wake the unconceived node?). This appendix states exactly what each check computes, derives both from the single objective the paper already carries, and lists the modelling commitments that remain after the derivation. The aim is that nothing in the two checks should read as a stipulated event: every acceptance is the model's own evidence weighed on the model's own ledger, and everything that is *not* derived is named as a commitment.

D.1 One ledger for every structural move

The bounded agent of §3.2 revises its beliefs by minimizing $D_{\mathrm{KL}}[q \,\|\, p(x \mid o, G)] - \lambda\, \mathbb{E}_q\big[U^\top x\big]$ (Eq. 3): fidelity to the honest posterior, traded against the value of where the commitments land, at temperature λ. A structural move—pruning an edge, waking a node—changes both terms, so the same objective scores the move:

$$\mathrm{score}(M \to \tilde{M}) \;=\; \underbrace{\Delta F}_{\text{log Bayes factor}} \;+\; \underbrace{\Delta G}_{\text{epistemic gain}} \;+\lambda\; \underbrace{\Delta U}_{\text{conviction value}}, \qquad \text{accept} \iff \text{score} > 0, \tag{9}$$

with ΔF the evidence for the edited structure, ΔG the expected information gain about any newly introduced latent (zero for a prune), and $\Delta U = \mathbb{E}_{q_{\tilde{M}}}\left[U^\top x\right] - \mathbb{E}_{q_M}\left[U^\top x\right]$ the change in the conviction-field value of the commitments. Equation (9) is not a new mechanism: it is Eq. (3) evaluated at two structures and differenced. Both threshold events of the simulation are instances of it.

D.2 The crisis check is the ledger on a prune

For a prune, every term of Eq. (9) is closed-form in the linear-Gaussian setting. The evidence half is Bayesian model reduction (Eq. 5): writing the agent's accumulated likelihood deposit as $J = \Pi - \Pi_0$, $j = h - h_0$, the Savage–Dickey identity scores the swap of the prior (Π_0, h_0) for the edge-e-reduced prior $(\Pi_0^{(e)}, h_0^{(e)})$ under the same likelihood, and—this is the part the main text does not spell out—it *already exhibits the reduced posterior*,

$$\Pi^{(e)} = \Pi_0^{(e)} + J, \qquad h^{(e)} = h_0^{(e)} + j, \tag{10}$$

so the value half costs one linear solve per candidate edge:

$$\Delta U_e \;=\; U^\top\big(\mu^{(e)} - \mu\big), \qquad \mu^{(e)} = \big(\Pi^{(e)}\big)^{-1} h^{(e)}, \quad \mu = \Pi^{-1} h, \tag{11}$$

with $U = \mathsf{T}u$ the agent's *current* conviction field, read along its present couplings. The crisis check is then exactly the ledger: edge e is flagged iff $\Delta F_e + \lambda\,\Delta U_e > 0$, and *crisis* is the round at which both protective belt edges are flagged, upon which the agent adopts the reduced prior composed over every edge its own ledger flags (the hub released). Pruning a conviction-protective edge moves the posterior mean away from the cherished commitments, so $\Delta U_e < 0$ and the rule reproduces the intuitive form "the evidence must overcome λ times the value the edge protects."

A natural surrogate suggests itself here: the static threshold $\Delta F_e > \lambda\, v_e$ with $v_e = |U_{\text{parent}}| + |U_{\text{child}}|$, the summed conviction magnitude at the edge's endpoints—available without a solve, and with the right *form* (it is Eq. (9) with $\Delta U_e \equiv -v_e$). But it has the wrong *content*: v_e is a static stand-in for a quantity the model can compute exactly, so we compute it. The difference is not cosmetic, and its direction is itself informative: because the reduced and full posteriors share the likelihood deposit, the mean displacement of a prune—and with it $\lambda\,\Delta U_e$—*shrinks as evidence accumulates*. A data-rich paradigm is barely protected at the threshold; almost all of its persistence must come from upstream, from the conviction gate that decides which evidence is allowed to accumulate at all (Eq. 4). The static surrogate would quietly overstate threshold protection; the exact ledger locates lock-in where the paper's results put it: gating plus isolation, not a high bar at the moment of decision.

D.3 The discovery check is the ledger on an expansion

Expansion cannot be closed-form (§3.3): a new node needs a real inversion, and—Stanford's problem of unconceived alternatives—the agent cannot plan toward a dimension it has not conceived. The discovery check therefore has exactly two parts, and only two:

1. **Arrival.** Candidate proposals arrive at a Poisson rate r. This is the one place a stipulated process enters, and it is a deliberate modelling commitment: a rate is the maximally agnostic model of conceiving what one has not conceived. The proposal's *direction* is read from the model's own prediction errors: a hidden hub b wired to the visible nodes with couplings c leaves, on marginalization, the rank-one footprint $-cc^{\top}/\Pi_{bb}$, so the leading eigenvector of the windowed error second moment is the best single-hub explanation of the residual. The window is an estimator horizon, not a gate.
2. **Acceptance.** The arrived proposal is accepted iff the *model log Bayes factor* of the bordered (hub-wired) model against the agent's current one is positive—that is, iff the residual genuinely loads on the proposed hub, equivalently iff the immediate Savage–Dickey prune of the new couplings would be rejected. The epistemic gain ΔG sits on the ledger too, but on the other side of the move: being positive for *any* new latent, it is what pays for attempting the (in general expensive) inversion, and it cannot keep a hub the data do not hold—an accept test that included it would wake on pure noise. Entertaining is paid by curiosity; keeping is paid by evidence. One consequence is owned rather than hidden: repeated looks give the Bayes factor a small false-discovery rate (rare noise crossings of $\Delta F > 0$ in the null world). We regard this as a feature of the honest model—communities do occasionally adopt spurious structure—and its principled remedy is not a reinstated threshold but the wake-then-prune cycle named in the limitations, in which a spuriously woken node is pruned back by the same reduction that scored it.

A tempting third part is deliberately absent: a *trigger*, requiring the residual floor (the eigenvalue above) to clear some height, perhaps for a sustained number of steps, before the Bayes factor is consulted. Such a gate would duplicate the accept test's job with a free constant—the Bayes factor already rejects proposals the residual does not support—and its only principled reading, a budget on when to pay for the expensive inversion, is a statement about real scientific communities rather than about this simulation, where the inversion is a small log-determinant. The discovery check therefore has no tuned threshold: the Poisson rate carries Stanford's commitment, the eigenvector carries the direction, and the ledger decides. (The floor is still recorded as telemetry, because it is the visible signature of the residual the old paradigm had been absorbing.)

D.4 What remains a modelling commitment

The derivation above leaves a short, explicit list of choices that are *not* forced by the formalism. We state them because the paper's claim that "crisis, revolution and discovery are threshold crossings of the model's own evidence" is honest only if its boundary is drawn:

- **The belt conjunction.** "Crisis" is operationalized as *both* named mass-law edges flagged in the same round. The per-edge accept is derived; the conjunction over those two edges is our operational definition of having eliminated the phlogiston mass-law reading.
- **The ordering — a result, not a commitment.** The simulation attempts expansion only after an agent's own crisis, but the ordering is enforced by the evidence rather than by fiat: running the identical proposal + ledger pipeline for every pre-crisis agent at every step (without applying it), the ledger rejects 13,180 of 13,185 pre-crisis agent-steps; the five crossings are marginal ($\Delta F < 0.005$, two orders below the post-crisis acceptance evidence) and are exactly the repeated-look false-positive rate owned above, whose remedy is the wake-then-prune cycle (§D.5). The intact paradigm absorbs the residual; there is nothing for the check to find until reduction exposes it.
- **The Poisson rate** r (swept), the prediction-error *window* (an estimator horizon), a short *warm-up* before reduction is permitted (an initialization guard), and the released hub's self-precision are quantitative choices of the same standing as the other dials of Table 1.
- **The closed-loop mechanisms' own commitments.** The mechanisms of Appendix E replace the constant rate, the one-shot accept and the static u with inferred quantities, but import quantitative choices of their own: the Hawkes kernel's parametric form (β, τ, exponential memory)—the *reading* of the kernel as filtered social evidence is motivated below, its shape is chosen; the expiring credit's geometric decay λ_c, a fixed-form surrogate for recomputing ΔG on the current posterior; and the audit's grace and patience horizons. All are Table 1-class dials, swept or defaulted, never tuned against an outcome.

D.5 The rate, made principled (and the audit that pays its debts)

This subsection records the inferential readings that license the closed-loop mechanisms whose quantitative results Appendix E reports: the adequacy-driven rate, the social excitation, the audit that the full ledger's acceptance rule makes necessary, and the accretion dynamics behind the Lakatos result—together with what each leaves stipulated.

The adequacy belief, filtered. One explores at the rate one believes one's model is incomplete, and that belief has a licensed likelihood: the residual floor the discovery check already records—the prediction error that persists after every within-model update, whose persistence is therefore evidence about the model *class* rather than its parameters. The agent tracks it with a fixed-gain filter, $s \leftarrow (1-\rho)\, s + \rho \cdot \text{floor}$, the same exponential-forgetting form as ω (and the fixed-gain limit of conjugate Gamma–Poisson updating under a drift prior on the latent rate), and sets $r(t) = r_0 + \kappa_r \max(s - s_{\text{ref}}, 0)$, capped at $r_{\max}$. A persistent residual raises the agent's own arrival rate; a quiet model lets it decay to the agnostic base. The clean waiting-time law of Appendix B does, as anticipated, dissolve into a feedback system; the experiments therefore report realized (effective) rates, and the matched-mean control of Appendix E is what separates "rate when needed" from "more rate."

The social stream, and why its filter is a Hawkes kernel. Because the agents share one hidden world, model inadequacy has a common cause across the community, and a trusted neighbour's discovery is evidence about one's *own* adequacy variable—the situation in which federated inference prescribes pooling [47]. Filtering the neighbours' discovery events (a point process, observed through the trust weights T_{ij} the population already fuses over) under an exponentially-correlated prior on the latent inadequacy yields a posterior intensity that is a shot-noise process: a baseline plus a decaying bump per observed event. That is exactly the linear self-exciting form of Hawkes [12],

$$r_i(t) \;=\; r_0 \;+\; \sum_{j \in \partial i} T_{ij} \sum_{t_j < t} \beta\, e^{-(t-t_j)/\tau}, \tag{12}$$

with t_j the neighbours' discovery times: β the trust-scaled evidential weight of one observed discovery, τ the assumed correlation time of inadequacy. The structural-proposal channel the model already has is thereby promoted from carrying *content* (which node to wake) to also carrying *momentum* (how eagerly to look)—a formal home for the observation that revolutions arrive in waves—and the branching ratio becomes the model's measurable answer to how much social excitement a revolution needs (Appendix E, Fig. 14). What the reading does not discharge: the kernel's parametric form is chosen, and the excitation deposited by a wake that is later pruned back (below) is not retracted—the community's excitement is an observation of the wake, not of the verdict.

The wake-then-prune audit. Accepting wakes on the full ledger $\Delta F + \Delta G > 0$ lets curiosity adopt structure the data do not (yet) hold, so the acceptance rule itself creates the obligation to audit. The audit re-asks the wake's own question every round: the bordered-model log Bayes factor of the agent's *current* (conditioned) beliefs against its *pre-wake* anchor at the stored coupling —not a Savage–Dickey prune of the wired prior, which would compare two priors and find the hub favoured by construction, since the slot has no observation row and is held only by the data's mean displacement along its pattern. The k-th audit's verdict is

$$\text{verdict}_k \;=\; \Delta F_{\text{now}} \;+\; \Delta G\, \lambda_c^{k}, \tag{13}$$

the wake's ledger with *expiring* credit: epistemic gain pays for entertaining, but it is a diminishing resource—information gain about a latent shrinks as the posterior over it sharpens—so the geometric λ_c is a fixed-form surrogate whose principled successor is recomputing $\varDelta G$ on the current hub posterior. A patience-length run of non-positive verdicts (after a grace period covering post-wake consolidation) re-pins the slot to its exact dormant form, and the agent may wake it again later: entertaining is paid by expiring curiosity, keeping by evidence. Figure 11 shows the cycle doing the false-discovery work the commitments list above assigns it: in a null world that never shifts, the credit wakes hubs in every agent and the audit re-pins every one of them; in the flip world the genuine hub is woken everywhere and never pruned.

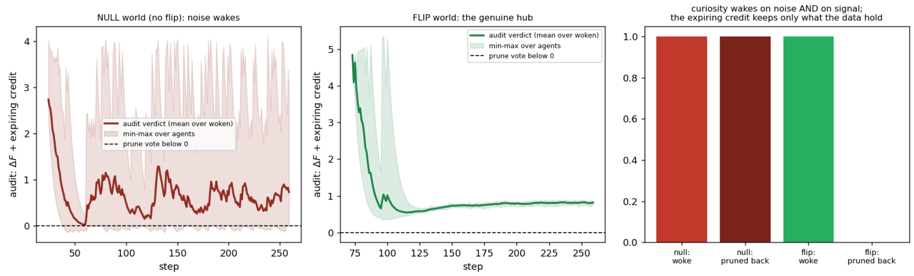

Fig. 11: The wake-then-prune cycle controls the repeated-look false-discovery rate with no threshold. *Left:* null world (no regime shift)—the audit verdict (evidence plus expiring credit) of every noise-woken hub decays below zero and the slot is re-pinned exactly. *Centre:* flip world—the genuine hub's verdict stays positive; the data take over before the credit runs out. *Right:* curiosity wakes structure in both worlds; the audit keeps only what the data hold.

Value accretion (the Lakatos closure). The conviction source u, exogenous and static in the baseline, gains slow dynamics: per node, $u_{\text{eff}} = u\,(1+g)$ with $g \leftarrow \text{clip}(g + \epsilon\, e - \delta\, g,\, 0,\, g_{\max})$ and e the agent's own entrenchment (the diagonal of its accumulated evidence deposit $\varPi - \varPi_0$, max-normalized)— self-evidencing made slow [13]: value flows toward the commitments the programme's own data have entrenched, and through $U = \mathsf{T}u$ and the gate of Eq. (4) a maturing programme increasingly silences its own disconfirming channels. The clip bounds the accretion–gate feedback loop. Figure 12 shows the resulting phase picture: gain accreting over the run, the realized gate climbing with ϵ, and conversion at the lock-in boundary falling—progressive and degenerating problemshifts as dynamics (Appendix E).

E Closed-loop mechanisms: quantitative results

The baseline model holds four quantities fixed that admit treatment as beliefs or protocols: the proposal rate, the acceptance of a wake, the conviction source u,

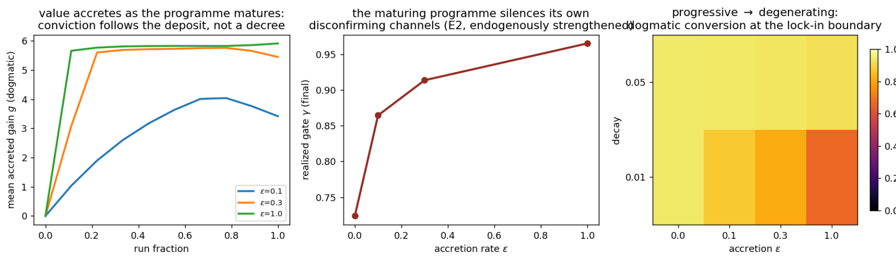

Fig. 12: Dynamic conviction (Lakatos, run forward). *Left:* the accreted gain g grows as the programme matures—conviction follows the agent's own evidence deposit, not a decree. *Centre:* the realized self-silencing gate γ strengthens with the accretion rate ϵ. *Right:* dogmatic-community conversion at the lock-in boundary over the (ϵ, decay) plane: a programme that matures into conservatism delays its supersession—and still loses to connection.

and the pooling rule. The inferential readings that license each mechanism are derived in Appendix D.5; this appendix reports what each does to the population results. Every mechanism defaults to "off"—constant rate, one-shot accept, static u, delta pooling—reproducing the baseline exactly, and each comparison below is run against that baseline with its own control.

The rate as a posterior over model adequacy. The agent tracks its residual floor—the leading eigenvalue of the windowed prediction-error second moment, the error no within-model update can absorb—with a fixed-gain filter $s \leftarrow (1-\rho)\,s+\rho{\cdot}\text{floor}$ and reads its proposal rate off the posterior, $r(t) = r_0 + \kappa_r \max(s - s_{\text{ref}}, 0)$. The result (Fig. 13) has the signature a posterior should have and a dial cannot: flat at r_0 through normal science, surging after the agent's own crisis exposes the residual (to an effective 0.34 from $r_0 = 0.01$), and decaying once the woken node explains it. The crisis-to-discovery delay falls from 55 to 29 steps—but the control that matters is the fixed rate it *matches*: a constant-rate agent given the adaptive run's realized mean rate achieves the same delay (27 steps). The gain is not more exploration; it is exploration when the agent's own evidence calls for it. Stanford's constraint survives: only the *propensity to look* is inferred, never a direction.

Social excitation rescues the staggered discovery. Filtering neighbours' discovery events through the trust weights, under an exponentially-correlated prior on the latent inadequacy, yields a posterior intensity of exactly the linear self-exciting (Hawkes) form [12],

$$r_i(t) \;=\; r_0 \;+\; \sum_{j\in\partial i} T_{ij} \sum_{t_j<t} \beta\, e^{-(t-t_j)/\tau}, \tag{14}$$

with t_j the neighbours' discovery times, β the trust-scaled evidential weight of one observed discovery, and τ the assumed correlation time of inadequacy (Ap-

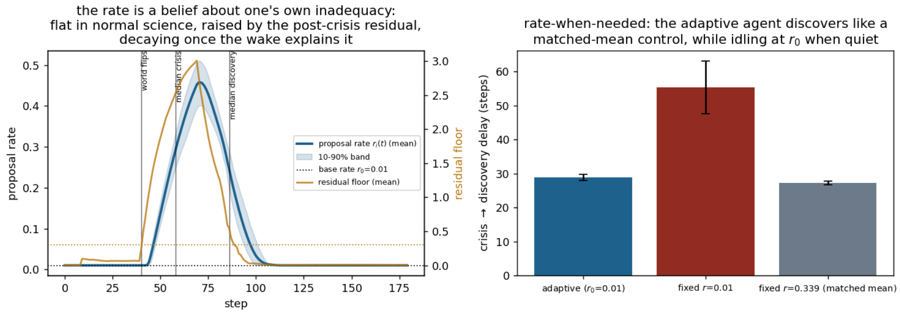

Fig. 13: The proposal rate as a posterior over model adequacy. *Left:* the per-agent rate (blue, mean and 10–90% band) is flat at the base rate r_0 through normal science, rises with the post-crisis residual floor (amber, right axis)—the evidence of model-class inadequacy the within-model updates cannot absorb—and decays once the woken node explains it. *Right:* crisis-to-discovery delay. The adaptive agent (left bar) matches a fixed-rate control given its realized mean rate (right bar) while idling at r_0 when the model is adequate: the gain is rate-when-needed, not more rate.

pendix D.5). Under Eq. (14) the first discovery raises the whole neighbourhood's propensity to look; the cascade compresses the community's wakings from a dispersion of 64 steps to 2, and—because survival is *completeness*-driven (a single still-pinned peer holds the fused slot precision enormous for every neighbour)—the completed cascade lets the anchored couplings consolidate. Survival rises monotonically in β from exactly zero at $\beta = 0$ to 0.88–0.95 at $\beta = 1.5$ across base rates $r_0 \in \{0.005, 0.01, 0.02\}$ at which an unexcited community essentially never completes (Fig. 14; 32 seeds per cell). The branching ratio of Eq. (14) is the model's quantitative answer to how much social excitement a revolution needs. One qualifier: survival is completeness-driven and therefore horizon-bound—a constant-rate community given a high enough rate and a long enough horizon also completes its cascade; what excitation adds is the mechanism by which a community reaches near-simultaneity at rates serendipity alone essentially never does.

The audited wake-then-prune cycle. Accepting wakes on the full ledger $\Delta F + \Delta G > 0$ lets curiosity adopt structure the data do not yet hold, and the audit of Appendix D.5 pays that debt: in a null world that never shifts, curiosity wakes structure in 100% of agents and the audit re-pins 100% of those wakes, exactly; in the flip world the genuine hub is woken everywhere and never pruned (Fig. 11). Two companion measurements close two more commitments. Widening the search—three eigen-hub candidates, each with its own dormant slot, plus the three largest residual couplings, 1,140 candidates priced on the same ledger—does not dilute discovery: the true cause wins 53 of 60 acceptances (median cosine 0.999 against the rank-one baseline). And running the identical discovery

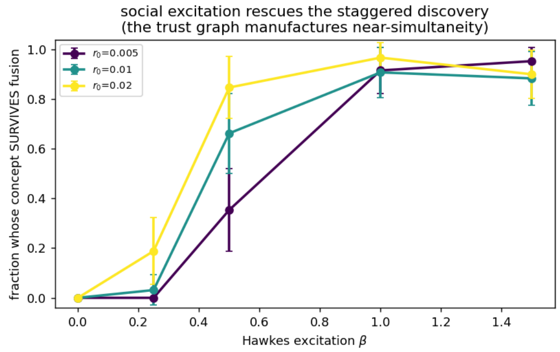

Fig. 14: Social excitation rescues the staggered discovery (sweep panel E, 32 seeds per cell). Fraction of the open community whose oxygen concept survives fusion, versus the excitation β of Eq. (14), at three base rates. At $\beta = 0$ (the constant rate) survival is exactly zero: staggered wakings are crushed one at a time. Excitation compresses the community's wakings in time, the cascade completes, no pinned peer remains in the fusion average, and survival rises to 0.88–0.95—the near-simultaneity that survival requires, manufactured by the community itself.

check *before* each agent's crisis, at every step, the ledger rejects 13,180 of 13,185 pre-crisis agent-steps: the crisis-before-expansion ordering is enforced by the evidence, not by fiat—the intact paradigm absorbs the residual, so there is nothing for the check to find until reduction exposes it.

Value accretion (Lakatos, run forward). Letting u accrete onto the commitments the agent's own evidence deposit has entrenched (Appendix D.5) closes the loop through the gate of Eq. (4): a maturing programme increasingly silences its own disconfirming channels (γ climbs from 0.72 to 0.97 as ϵ rises), conversion at the lock-in boundary falls from 0.94 to 0.67, and at very weak coupling the dogmatic community's conversion is *delayed* roughly 2.2-fold rather than prevented (Fig. 12). Conviction that grows from the inside still loses to connection: it buys the old paradigm time, not immunity—a fair gloss of a degenerating problemshift.

The mechanisms compose. With the adequacy-posterior rate, social excitation, the expiring-credit audit, abstention pooling and value accretion all enabled in one population, the full arc still runs—crisis at $t \approx 67$, discovery at $t \approx 69$, the rate surging $0.02 \to 0.77$ and decaying back to base, zero false prunebacks, concept survival complete, and the never-shifting null world silent in every channel. No mechanism breaks another, because none adds a new objective: each routes a quantity the ledger already computes—residual, epistemic gain, evidence, deposit—back into a parameter that had been a constant.